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RESEARCH ARTICLE

DNN-VTRM Channel Estimation for OFDM Underwater Acoustic Communication System

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ABSTRACT Nonlinear factors can adversely affect underwater acoustic (UWA) channel estimation, and the multipath and large delay extension of the UWA channel still causes residual inter-symbol interference (ISI) for orthogonal frequency division multiplexing (OFDM) underwater acoustic communication systems. In this study, the Deep Neural Network (DNN) and Virtual Time Reversal Mirror (VTRM) technologies are jointly used as channel estimation and equalization algorithms in the OFDM underwater acoustic communication system. VTRM technology is used to solve the issues of OFDM system performance degradation caused by the residual ISI. The DNN network is used to solve the nonlinear factor from the UWA channel and the floor effect of the LFM matched filtering algorithm in the channel estimation part of the VTRM, so that it can complete the UWA channel mapping better, including sparsity, large delay expansion, and multipath. First, the DNN time-domain channel estimator is trained using the received random probe signal and sparse UWA multipath channel impulse responses generated by Bellhop. The output value of the DNN channel estimator is then processed by the filtering algorithm to restore the sparsity of the channel, which prepares as the UWA channel impulse response. When the received signal is preprocessed by VTRM equalization, the focusing principle of time reversal can be used to cancel the distortion caused by the UWA sparse multipath channel. Finally, this study uses the BELLHOP model to simulate the UWA test channel to compare the DNN-VTRM with traditional channel estimation methods such as the Least Square (LS) algorithm, the Linear Minimum Mean Square Error (LMMSE) algorithm, the Orthogonal Matching Pursuit (OMP) algorithm, and the DNN frequency domain channel estimator. The experimental results show that the estimated effect of the DNN time-domain channel estimator is more accurate than that of the LFM matched filtering and OMP algorithms. Under the low signal-to-noise ratio (SNR) condition, the OFDM underwater acoustic communication system based on DNN-VTRM has a better bit error rate (BER) performance than the traditional algorithm.

INDEX TERMS Underwater acoustic communication, orthogonal frequency division multiplexing, deep neural network, virtual time reversal mirror, channel estimation.

I. INTRODUCTION

Orthogonal Frequency Division Multiplexing (OFDM) as a high-data-rate communication solution is attracting

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increasing attention owing to its capability to combat multipath effects and its high spectral efficiency. However, the severe Doppler shifts, high noise, strong multipath, and time-varying nature of underwater acoustic (UWA) channels still result in varying degrees of inter-symbol interference (ISI) in OFDM systems. In a shallow sea environment, the

distribution of the UWA channel sound speed will cause a serious multipath, and the multipath delay is more than tens of milliseconds. At the same time, acoustic signals are also affected by variations in the marine environment, including water velocity, density, and temperature gradient [1]. The acoustic signal undergoes absorption and scattering in the process of propagation, which weakens the strength of the acoustic signal and prevents it from reaching the expected receiver with the communication distance increasing. Virtual Time Reversal Mirror (VTRM) technology is known for its effective multipath suppression and channel energy focusing capabilities, which are gradually applied to the OFDM underwater communication system. Accurate channel estimation remains a critical challenge for VTRM technology.

The traditional method uses the Linear Frequency Modulation (LFM) signal to complete the channel estimation part of the VTRM. However, owing to the accuracy issues of the LFM matched filtering algorithm, it has a floor effect in the channel estimation processing stage, in which the estimation accuracy performance cannot be further improved and always remains at a certain value. Aiming at the problem of the residual ISI caused by the multipath of the UWA channel for the OFDM underwater acoustic communication system, the nonlinear factor of the UWA channel, and the floor effect of the LFM matched filtering algorithm in the VTRM channel estimation, this study combines a Deep Neural Network (DNN) with VTRM technology for application in the OFDM underwater acoustic communication system, which can better adapt to the UWA channel under the unknown channel condition. This offers the advantage of solving the nonlinear interference factors in the UWA channel and the floor effect of the LFM matching filtering algorithm, which can further improve the UWA channel estimation and equalization performance of the OFDM system.

A. RELATED WORK

In recent years, researchers have attempted to apply Time Reversal Mirror (TRM) technology to OFDM underwater acoustic communication systems and have achieved some notable outcomes. Vadde et al. from the PES Institute of Technology, India, investigated the telemetry advantages of combining OFDM with Active Time Reversal Mirror (ATRM) technology from a channel perspective, and proposed a linear minimum mean square error channel estimation algorithm resembling a Wiener filter [2]. Liu and Yang from Sun Yat-Sen University, Taiwan, studied the impact of the OFDM cyclic prefix length on time reversal channel equalization [3]. Gomes and Barroso from Lisbon University, Portugal, streamline the design of communication protocols by using the TRM technology to achieve focused information in transmitter arrays [4]. Gomes et al. proposed a method using the Passive Time Reversal Mirror (PTRM) technology for multichannel combining of observed waveforms to produce an equivalent channel with a shorter impulse response. This approach can improve the spectral efficiency of OFDM

by reducing the need for pilot symbols [5]. Gomes et al. validated that the PTRM technology effectively reduces the OFDM cyclic prefix length, thereby improving the spectral efficiency [6]. Yin et al. from Harbin Engineering University applied the Virtual Time Reversal Mirror (VTRM) technology combined with the Matching Pursuit (MP) algorithm for the OFDM underwater acoustic communication system, and the results showed that the performance of VTRM channel equalization is better than that of the PTRM and the Least Square (LS) channel equalization [7]. Yin et al. proposed a virtual time reversal processing (VTRP) algorithm for a single-input single-output (SISO) OFDM system, and the results showed that VTRP has outstanding performance in time-invariant channels [8]. Zhang from Yanshan University applied fractional Fourier transform channel estimation technology and the Orthogonal Matching Pursuit (OMP) algorithm in OFDM underwater acoustic communication systems [9]. Sun from Harbin Engineering University studied multi-channel OFDM underwater acoustic communication systems based on spatial diversity and VTRM technology. Simulation results indicate that the VTRM-OFDM system can further enhance the performance of underwater acoustic communication compared to a single spatial diversity OFDM system [10]. Zhao et al. from Xiamen University designed a UWA voice communication system combining multi-channel VTRM and time-frequency differential OFDM, achieving temporal and spatial multipath focusing and suppression of residual multipath effects [11]. Numerous studies have advanced VTRM technology; however, the UWA channel is a complex and nonlinear variation, which is a challenge in the channel estimation part of VTRM technology.

Currently, some research achievements are applying neural networks to the UWA channel estimation, but the number of studies related to the UWA channel remains limited. In a study of three-dimensional imaging of the UWA sonar, Dongdong Zhao from Zhejiang University of Technology proposed a UWB-DPS algorithm. By optimizing the sparsity of the planar transducer array, the sparsity rate is as high as 97.7%. Experimental results show that the UWB-DPS algorithm based on FPGA has a faster computation speed and higher imaging quality [12]. Fu et al. from Tianjin University have constructed a passive time reversal convolutional neural network at the receiver end of OFDM underwater communication systems, validating its feasibility and robustness through field experimentation [13]. Jiang et al. from the Beijing Institute of Technology proposed two different DNN channel estimators and introduced a new training strategy, which demonstrated that the DNN could effectively learn nonlinear variations in the UWA channel [14]. Youwen Zhang, et al. from Harbin Engineering University utilized a sound propagation model with measured sound speed profiles to train and test a DNN receiver and proposed a DNN model for an OFDM underwater acoustic communication system. The simulation showed a significant performance improvement compared with traditional OFDM underwater acoustic receivers [15]. Senol et al. from Istanbul University

designed an Artificial Neural Network (ANN) to estimate the sparse multipath channel, demonstrating superior mean square error (MSE) performance over matching pursuit (MP) and orthogonal matching pursuit (OMP) algorithms [16]. Li et al. from the Beijing University of Technology proposed an OMPC channel estimation algorithm based on the BP network, addressing the challenges of the OMP algorithm, such as its high computational complexity, slow convergence, and ineffective utilization of channel sparsity in sparse channel estimation [17]. Sicong Liu, et al. from Xiamen University introduced a novel DL-UAC method, which combined the conventional iterative sparse recovery algorithm of approximate message passing (AMP) with a DNN to construct a sparsity-aware DNN for the deep learning of the inherent sparse structure of the UAC. The experimental results showed that DL-UACE outperformed the OMP and AMP algorithms regarding the estimation accuracy and spectral efficiency under a low signal-to-noise ratio (SNR) and insufficient pilot conditions [18]. Yunfei Li, et al. from the Anhui University of Technology proposed a variational Bayesian learning framework based on localization and channel reconstruction, which utilizes the sparsity of the angular domain to unify near field and distant field scenes. The computational complexity and estimation accuracy are superior to the iterative algorithm based on particle swarm optimization (PSO) and the iterative algorithm based on maximum likelihood (ML) [19]. For the holographic MIMO communication system, the traditional far-field-only and near-field-only channel estimation method may not be suitable for the hybrid near-field and far-field modes. Shaohua Yue from Peking University demonstrated the existence of power diffusion (PD) effects and proposed a PD-aware orthogonal matching pursuit (PD-OMP) algorithm. The experimental result shows that the PD-OMP algorithm can accurately estimate the channel without prior knowledge of the near-field and far-field information and the user location, which is better than the current most advanced the hybrid near-field and far-field channel estimation method [20]. In addition, the related literature regards the neural network as the receiving module part of the OFDM system. Yen and Chen employed a deep learning network to solve the problem of rapid changes in the UWA channel, the results revealed that the VTRM technology can generally mitigate the bit error rates (BER) of the deep learning network receiver and the traditional demodulation receiver [21]. To improve the performance under a low signal-to-noise ratio (SNR) and multipath channel condition, Liu et al. proposed a cyclic shift keying spread spectrum (CSK-SS) underwater acoustic communication system based on deep learning, and the experimental data showed that the preprocessing step using the VTRM technology can reduce the BER of the system [22]. Liu et al. proposed a novel approach for chirp spread spectrum (CSS) communication, and the result shows that the chirp spread spectrum 1D CNN and MLP receivers could get more gain by VTRM preprocessing [23].

In summary, the VTRM technology and neural network can improve the MSE and BER performance of the OFDM underwater acoustic system; therefore, this paper proposes to reduce the residual ISI further by using the DNN-VTRM at the receiver of the OFDM communication system. However, the equalization performance of the VTRM depends on the accuracy of the UWA channel estimation. In this study, the DNN model is used as the channel estimation part of the VTRM, which overcomes the floor effect of the LFM matched filtering algorithm as the VTRM channel estimation part and improves the adaptability of the OFDM system to the nonlinear variation of the UWA channel.

B. OUR CONTRIBUTION

In the content arrangement of this paper, Section I is the introduction part, which mainly analyzes the related research status and summarizes the main research content of this paper. Section II introduces the underwater communication OFDM system based on DNN-VTRM, the DNN-VTRM module and its processing flow, the DNN structure setting, and the DNN model training. Section III introduces the UWA environment and the DNN-VTRM communication system parameter setting. Section IV introduces and analyzes the UWA virtual time inverse channel. Under the 16QAM and 64QAM modulation, it uses comparative analysis the mean square error (MSE) and the bit error rate (BER) among the LFM matched filtering algorithm, the OMP algorithm, and the DNN time domain estimator. Simultaneously, this section compares the DNN-VTRM system with the traditional LS algorithm, LMMSE algorithm, and the DNN frequency domain estimator without the VTRM processing to analyze the BER performance of these methods. Section V summarizes the study.

The primary contribution of our work can be summarized as follows:

- 1) To improve the reliability and accuracy of the experiment, this study used the BELLHOP ray model to simulate the UWA channel environment and to generate different training data samples. The multipath number, delay, complex gain, and phase of each training sample is completely different.
- 2) This research demonstrates the influence of the VTRM technology on the UWA channel and the OFDM underwater acoustic communication system. Meanwhile, it explains that the cyclic prefix and the cyclic suffix can help the OFDM underwater acoustic communication system to overcome the ISI from the sidelobe of the UWA time-inverse channel.
- 3) We use the DNN time domain channel estimator to obtain the UWA test channel information including the delay, complex gain, and phase. Then, we compare the LFM matched filtering algorithm, the OMP algorithm, and the DNN time domain channel estimator from the MSE and BER views.

- 4) We use the DNN-VTRM system to compare the BER performance with the conventional LS, LMMSE, and the DNN frequency domain channel estimator without the VTRM processing under the 16QAM and 64QAM modulation. Moreover, we analyze the system's performance under the low and high SNR conditions.
- 5) We compare the efficiency of the LFM-VTRM, the OMP-VTRM, the DNN-VTRM, the LS algorithm, the LMMSE algorithm, and the DNN frequency domain estimator without VTRM processing by computing the product complexity and running times.

II. OFDM UWA COMMUNICATION SYSTEM BASED ON DNN-VTRM

A. SYSTEM MODEL

Fig. 1 shows the architecture of an OFDM underwater acoustic communication system based on DNN-VTRM. At the transmitter, a binary bit stream from the information source undergoes modulation and encoding to generate the complex data symbol $S(k)$, which is converted from serial to parallel according to the number of subcarriers in the system. The parallel data is used to generate information symbols $s(n)$ through the Inverse Fast Fourier Transform (IFFT) with N subcarriers. To obtain the time-domain signal $s(n)$ from the frequency-domain signal $S(k)$, the following formula is used:

$$\begin{aligned} s(n) &= \text{IFFT}\{S(k)\} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} S(k) e^{j\frac{2\pi}{N}nk}, \quad n = 0, 1, \dots, N-1. \end{aligned} \quad (1)$$

A cyclic prefix and suffix are added to $s(n)$, which is parallel to serial conversion. Before transmission, LFM and a probe signal $p(n)$ are added to form the transmission data frame, as shown in Fig. 2. The $p(n)$ inputs the DNN network to complete the UWA channel estimation for the entire DNN-VTRM system. Simultaneously, the LFM signal is also required to complete the matched filtering algorithm as a contrast experiment. Each signal has a guard interval that exceeds or equals the maximum UWA channel delay. The received information signal $s_r(n)$ through the UWA channel $h(n)$ can be represented by the following equation:

$$s_r(n) = s(n) \otimes h(n) + g(n), \quad (2)$$

$h(n)$ and $g(n)$ denote the UWA channel impulse response and zero-mean additive white Gaussian noise, respectively. $\otimes$ denotes a circular convolution. The UWA channel $h(n)$ can be expressed by equation (3) as follows:

$$h(n) = \sum_{l=0}^{L-1} A(l) \delta(n-l), \quad (3)$$

where L denotes the length of the UWA channel. $A(l)$ represents the impulse response amplitude at the l -th position of the UWA channel. $\delta(n-l)$ represents the unit impulse response at the l -th position of the UWA channel.

The VTRM theory can effectively mitigate the residual ISI of the OFDM system caused by the UWA multipath channel. At the receiver, the received signal $s_r(n)$ is performed the virtual time reversal processing to obtain the information signal $r(n)$, as shown in Fig. 3. The detailed processing steps are introduced in the DNN-VTRM model chapter. The accuracy of the UWA channel estimation plays a critical role in determining the final effect of the VTRM. In a traditional OFDM system, channel estimation typically occurs after a Fast Fourier Transform (FFT) by using pilot data. However, in this study, the channel estimation precedes the FFT stage, which is aligned with the technical characteristics of the VTRM. Strictly speaking, the channel estimation should be integrated into a VTRM module. However, a separate channel estimation is essential for coherent demodulation in an OFDM system. Therefore, the channel estimation should serve as a shared module for the entire system.

According to the flowchart in Fig. 1, when the system completes the virtual time reversal processing, serial to parallel conversion and removes the cyclic prefix and suffix. The time domain signal $r(n)$ is transformed into the frequency domain signal $R(k)$ by the FFT stage. The stage in equation (4) is as follows:

$$\begin{aligned} R(k) &= \text{FFT}\{r(n)\} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} r(n) e^{-j\frac{2\pi}{N}nk}, \quad k = 0, 1, \dots, N-1. \end{aligned} \quad (4)$$

When the serial transformation of the frequency domain signal $R(k)$ is completed, signal $R(k)$ needs to be recovered, which requires the demodulation and decoding.

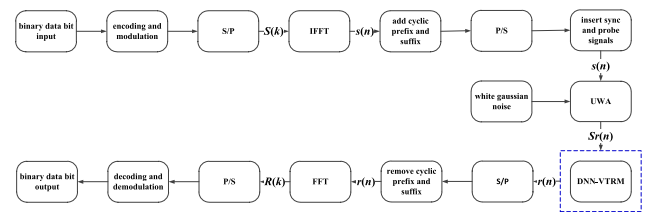


FIGURE 1. Underwater acoustic OFDM communication system based on DNN-VTRM.

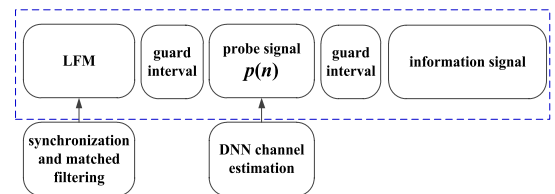


FIGURE 2. Transmitted signal frame structure.

B. DNN-VTRM MODEL

As shown in the Fig. 3, The fundamental principle of the DNN-VTRM algorithm need a data frame including the probe signal $p(n)$ and the signal $s(n)$ to pass through the UWA channel. At the receiving end, the received probe signal $p_r(n)$

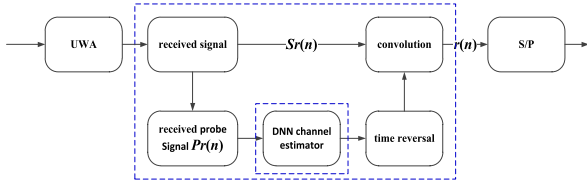


FIGURE 3. Structure of DNN-VTRM.

is used to estimate the UWA channel impulse response $\hat{h}(n)$. Subsequently, the received signal $s_r(n)$ need to convolve with the UWA time reversal channel estimate response $\hat{h}(-n)$ to obtain the signal $r(n)$.

The specific calculation formula is as follows:

$$r(n) = s_r(n) \otimes \hat{h}(-n). \quad (5)$$

Substituting equation (2) into equation (5), the following equation is obtained:

$$\begin{aligned} r(n) &= s(n) \otimes h(n) \otimes \hat{h}(-n) + g(n) \otimes \hat{h}(n) \\ &= s(n) \otimes h'(n) + g(n) \otimes \hat{h}(n), \end{aligned} \quad (6)$$

where, $h'(n) = h(n) \otimes \hat{h}(-n)$ represents the virtual time reversal channel, $h'(n)$ is the cross-correlation of the $h(n)$ and the $\hat{h}(n)$. If the $\hat{h}(n) = h(n)$, then $h'(n)$ is an autocorrelation of the $\hat{h}(n)$. The first term on the right-hand side of equation (6) represents useful information after the signal passes through the virtual time reversed channel, whereas the second term on the right-hand side represents the additive time reversed noise.

In formula (3), the discrete channel $h(n)$ of length L is expressed as:

$$h(n) = [h(0) \ h(1) \ \cdots \ h(L-1)]. \quad (7)$$

Among these, the $h(0)$ is a direct sound with the largest amplitude. The virtual time-inverse channel through equation (6) can be expressed as:

$$h'(n) = [h'(1-L) \ \cdots \ h'(-1)h'(0)h'(1) \ \cdots \ h'(L-1)]. \quad (8)$$

In this case, the length of the $h'(n)$ is $2L-1$, but the channel taps $h'(1-L), \dots, h'(-1)$ and $h'(1), \dots, h'(L-1)$ have symmetry and similarity, which is called the side lobe. The $h'(0)$ represents the superposition of each multipath, called the main lobe. The amplitude of the main lobe is significantly higher than that of the side lobe, which can help to reduce the residual ISI caused by the multipath expansion and to improve the system gain at the low SNR.

Among them, a side lobe with a small amplitude is regarded as noise, which can be ignored. By the VTRM processing, the amplitude ratio of the side lobe to the main lobe of the $h'(n)$ is much lower than that of the original channel $h(n)$, thus compressing the time length of the UWA channel.

If the $h(n)$ is a time-invariant channel in the OFDM symbol period, and the DNN channel estimator can accurately estimate the $\hat{h}(n) = h(n)$, there will be the equation $h'(n) = \hat{h}(n) \otimes \hat{h}(-n)$. As shown in Figs. 9(a) and 9(b), the $h'(n)$

still has some side lobes by the VTRM processing. It is necessary to add a cyclic prefix and cyclic suffix to overcome the interference of the side lobe. At this time, the virtual time reversal channel $h'(n) \approx \delta(n)$, where $\delta(n)$ denotes a unit impulse response [24]. The signal $s(n)$ by the VTRM processing is given by the following formula:

$$s(n) \otimes h'(n) \approx s(n) \otimes \delta(n) = s(n),$$

$$r(n) \approx s(n) + g(n) \otimes \hat{h}(-n). \quad (9)$$

Processed using equation (9), $s(n)$ and $r(n)$ exhibit similar waveforms, where the noise term $g(n) \otimes \hat{h}(-n)$ represents the convolution of the received noise and the estimated time reversed channel. In this case, the received signal $r(n)$ is only disturbed by the additive Gaussian noise. The following is a detailed description for overcoming the interference from the $h'(n)$ side lobe by the cyclic prefix and cyclic suffix.

When the m -th symbol is s^m , there is not only the delay spread of the previous symbol s^{m-1} , but also the next symbol s^{m+1} will influence the current symbol s^m [7]. If the cyclic prefix and suffix prefix are not added, after passing through the UWA channel, the corresponding receiving symbol can be expressed as:

$$\mathbf{r}^m = \mathbf{h}^m \mathbf{s}^m + \mathbf{h}^{m-1} \mathbf{s}^{m-1} + \mathbf{h}^{m+1} \mathbf{s}^{m+1} + \mathbf{n}^m, \quad (10)$$

where, the $\mathbf{r}^m, \mathbf{s}^m, \mathbf{s}^{m-1}, \mathbf{s}^{m+1}$ and $\mathbf{n}^m$ represents the $N \times 1$ dimensional column vectors, which should need to satisfy the $N \geq L$ condition. The L represents the UWA channel $h(n)$ length.

$\mathbf{h}^m$ can be represented as:

$$\mathbf{h}^m = \begin{pmatrix} h'(0) & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ h'(L-1) & \cdots & h'(1) & h'(0) & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & h'(L-1) & h'(L-2) & \cdots & h'(0) \end{pmatrix}, \quad (11)$$

$\mathbf{h}^{m-1}$ can be represented as:

$$\mathbf{h}^{m-1} = \begin{pmatrix} 0 & \cdots & h'(L-1) & h'(L-2) & \cdots & h'(1) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & h'(L-1) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix}, \quad (12)$$

$\mathbf{h}^{m+1}$ can be represented as:

$$\mathbf{h}^{m+1} = \begin{pmatrix} 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ h'(1-L) & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ h'(-1) & \cdots & h'(2-L) & h'(1-L) & \cdots & 0 \end{pmatrix}, \quad (13)$$

where, the $\mathbf{h}^m$, $\mathbf{h}^{m-1}$, and $\mathbf{h}^{m+1}$ represent the $N \times N$ dimensional matrix.

The $\mathbf{h}^{m-1}\mathbf{s}^{m-1}$ and $\mathbf{h}^{m+1}\mathbf{s}^{m+1}$ in equation (10) denote the interference of the previous symbol and the next symbol for the current symbol $\mathbf{h}^m\mathbf{s}^m$, respectively. And $\mathbf{n}^m$ is an additive noise. The cyclic prefix can eliminate the effect of the $\mathbf{h}^{m-1}\mathbf{s}^{m-1}$, and the cyclic suffix can eliminate the effect of the $\mathbf{h}^{m+1}\mathbf{s}^{m+1}$. If the length of the cyclic prefix and the cyclic suffix exceed the maximum length L of the $h(n)$, after removing the cyclic prefix and the cyclic suffix, the receiving symbol $\mathbf{r}^m$ can be expressed as:

$$\mathbf{r}^m = \tilde{\mathbf{h}}^m \mathbf{s}^m + \mathbf{n}^m. \quad (14)$$

The $\tilde{\mathbf{h}}^m$ represents a cyclic matrix, which can be expressed as:

$$\tilde{\mathbf{h}}^m = \begin{pmatrix} h'(0) & h'(-1) & \cdots & h'(2) & h'(1) \\ h'(1) & h'(0) & \ddots & h'(3) & h'(2) \\ \vdots & \ddots & \ddots & 0 & \ddots \\ h'(-1) & h'(-2) & \cdots & h'(1) & h'(0) \end{pmatrix}. \quad (15)$$

By comparing equation (10) and (14), we can see that the ISI caused by the previous symbol and the next symbol can be eliminated by the cyclic prefix and the cyclic suffix.

C. DNN NETWORK STRUCTURE

As is well known, the DNN inherit the structure of classical neural networks, which include more hidden layers and neurons. This deep architecture enables the model to perform better than the traditional methods in complex and highly nonlinear problems. Because the deep architecture can help to extract more unknown and potential feature information of the UWA channel, which traditional methods do not find. Therefore, we use the DNN channel estimator to estimate the VTRM channel responses, which has excellent learning, representation, and approximation capabilities for UWA channel estimation problems.

DNN networks usually consist of numerous neurons that are fully interconnected with each other to form a fully connected layer. Each pair of neurons in the adjacent layers is connected by a weight factor that is dynamically adjusted during training. This structure allows the output of the neurons from the previous layer to serve as the input to the neurons in the subsequent layer. The leftmost layer of the network is known as the input layer, and the rightmost layer is known as the output layer. Several intermediate layers between the input and output layers, whose values are not visible during training, are termed hidden layers, owing to their hidden characteristics. The DNN network structure adopted in this study is illustrated in Fig. 4:

For a DNN model, the first step is to select the suitable hyperparameters. It primarily involves defining the neural network structure, which includes determining the number of neurons in each layer and the number of the hidden layer. Next, we select the type of activation function for each layer (such as sigmoid, tansig, relu, and purelin functions) and the cost function (such as the mean squared error or

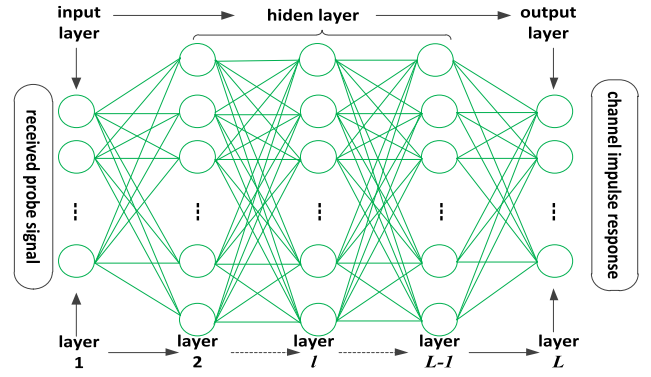


FIGURE 4. Structure of the Deep Neural Network.

cross-entropy). Finally, these hyperparameters determine the structure of the DNN. In this study, the tansig and purelin functions are used as activation functions for the hidden and output layers, respectively. The tansig and purelin functions are defined as follows:

$$f_1(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad (16)$$

$$f_2(x) = cx + d. \quad (17)$$

The output range of the tansig function is $[-1, 1]$ as shown in Fig. 5-(a). Meanwhile, the tansig function helps to alleviate the vanishing gradient problem, and its nonlinear characteristics can help extract more feature information about the UWA channel from complex datasets. As the activation function of the output layer, the purelin function is a linear variation that can ensure that no additional nonlinear transformation affects the output layer, as shown in Fig. 5-(b).

Assuming that the number of DNN layers is L , where $l \in [1, L]$ denotes the l layers. Thus, the first layer is the input layer, and the L -th layer is the output layer. Suppose that the input layer data vector can be represented as:

$$\mathbf{p}^{(1)} = [p_1^{(1)}, p_2^{(1)}, \dots, p_j^{(1)}, \dots, p_{J^{(1)}}^{(1)}]^T, \quad (18)$$

where $J^{(1)}$ represents the number of neurons in the first layer (the input layer), and T denotes the matrix transpose. Then, the input to the j -th neuron in the second layer is:

$$p_j^{(2)} = \sum_{i=1}^{J^{(1)}} w_{ij}^{(1)} p_i^{(1)} + b_j^{(1)}, j = 1, 2, \dots, J^{(2)}, \quad (19)$$

where $w_{ij}^{(1)}$ represents the weight between the i -th neuron in the first layer and the j -th neuron in the second layer, $b_j^{(1)}$ denotes the bias of the j -th neuron in the second layer, and $J^{(2)}$ denotes the number of neurons in the second layer. In each hidden layer ($l = 2, 3, \dots, L-1$), there exists a nonlinear activation function $f^{(l)}(\cdot)$, that transforms the linear combination of input into a nonlinear output. Therefore, the output of the j -th neuron in the second layer is:

$$q_j^{(2)} = f^{(2)}(p_j^{(2)}). \quad (20)$$

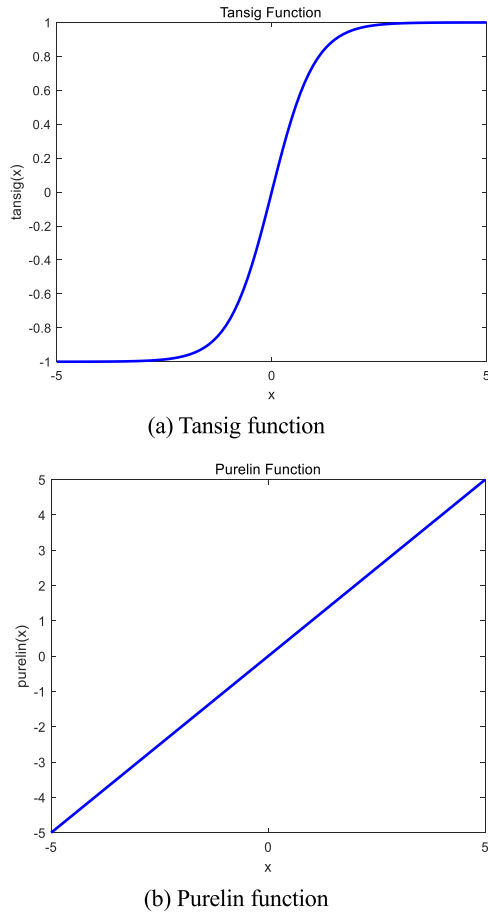


FIGURE 5. Activation functions in neural networks.

The output vector of the second layer can be represented as follows:

$$\mathbf{q}^{(2)} = [q_1^{(2)}, q_2^{(2)}, \dots, q_j^{(2)}, \dots, q_{J^{(2)}}^{(2)}]. \quad (21)$$

Equation (21) represents the input of the next layer. In this manner, the input $p_j^{(l)}$ and output $q_j^{(l)}$ of the j -th neuron in the l -th layer ($l = 2, 3, \dots, L$) are represented as follows:

$$p_j^{(l)} = \sum_{i=1}^{J^{(l-1)}} w_{ij}^{(l-1)} p_i^{(l-1)} + b_j^{(l-1)}, j = 1, 2, \dots, J^{(l)}, \quad (22)$$

$$q_j^{(l)} = f^{(l)}(p_j^{(l)}), \quad (23)$$

where $J^{(l)}$ denotes the number of neurons in the l -th layer of the network.

In an OFDM system, the time domain signal is expressed in a complex form, which consists of real and imaginary parts. Therefore, the neural network divides the complex form signal into real and imaginary components as the input. Within the same layer of the neural network structure, starting from the first neuron, every two adjacent neurons handle one complex data (real and imaginary parts) to input, which allows J neurons to simultaneously correspond to $J/2$ complex data. In this study, the received discrete probe signal $p_r(n)$ is used

as the input of the DNN network, and uses the random UWA channel datasets generated by the Bellhop as the training labels, as shown in Fig. 4. Hence, the number of neurons in the input layer must be twice the length of the received discrete probe signal $p_r(n)$ and the output layer should correspond to twice the maximum delay of the training sample.

Appropriately increasing the number of hidden layers and the number of neurons is conducive to the feature extraction of the UWA sparse channel, but too many neurons will cause the DNN model to overfit or reduce the generalization ability. The number of neurons in the hidden layers is determined by the maximum delay of the UWA sparse channel training sample. Data preprocessing can reduce the complexity of the data and increase the number of hidden layers to improve the accuracy of the DNN model. In this study, the number of neurons in the hidden layer is chosen to be twice the maximum time delay of the training sample. The specific parameters of the DNN structure are presented in table 1.

TABLE 1. DNN network structure.

Layer	Nodes	$f(\cdot)$
Input	128	—
Hidden #1	128	tansig
Hidden #2	128	tansig
Hidden #3	128	tansig
Output	128	purelin

D. DNN NETWORK TRAINING

Generally speaking, the deployment of a DNN model is divided into two phases: the training and testing stage. To estimate the UWA channel parameters effectively in the working stage, the DNN model must be trained with the OFDM modulated probe signal during the training process. The training process of the DNN model can be summarized as the forward propagation of the training symbols and the backward propagation of the learning errors. When the discrete probe signal $p(n)$ through the UWA channel to get the received probe signal $p_r(n)$, it will be affected by the nonlinear factor caused by multipath and large delay expansion. The received discrete probe signal $p_r(n)$ and the known UWA channel dataset are mainly used for model training, as shown in Fig. 6. In addition, the DNN output value will compare with the UWA channel labels generated by the Bellhop to calculate the learning error. If the channel estimation result of the DNN output layer does not match the known the UWA channel label, the learning error between the network output and the target output uses the back propagation (BP) algorithm to return from the output layer to the hidden layer to correct the weights and biases of these neurons [25]. If the error value is within the allowable range, a series of weights and bias parameters of the DNN model are saved, and DNN training is completed; otherwise, the training process is restarted. The

training process aims to minimize the difference between the target data sequence and the output sequence of the DNN model. Here, the Mean Squared Error (MSE) is selected to define the feedback information during the DNN training stage, and the cost function is as follows:

$$E = \frac{1}{N} \sum_{n=0}^{N-1} (h'' - h(n)), \quad (24)$$

where $h(n)$ represents the state of each UWA channel label in the training datasets, $\hat{h}(n)$ represents the output value of the DNN model for each UWA channel label, and the E value represents the MSE between $h(n)$ and $\hat{h}(n)$. In the training phase, if the E value of Equation (24) reaches a predefined threshold, appropriate weights and bias parameters are generated for each layer of the DNN model, which expresses the mapping of the UWA channel nonlinear factor. which is used by the DNN-VTRM system during the working phase. In addition, in order to improve the adaptability of the DNN network to environmental noise, the model is trained by mixing the probe signal under variety of SNRs.

In this paper, the Resilient Back Propagation (Rprop) training algorithm is used to train the DNN network, and this algorithm is a gradient batch update function based on the Manhattan update rule. The learning rate of each weight is independent and dynamically adjusted depending on the direction of the gradient rather than the magnitude of the gradient. The principles of the Rprop algorithm is presented in Table 2:

The conventional Levenberg Marquardt (LM) training algorithm must consider the direction and magnitude of the gradient. It computes the Hessian matrix constantly during the iterative process, which requires more time and memory [26]. The Rprop algorithm does not easily fall into a local minimum, requires less memory, and has a faster convergence rate. The experiment attempted to use the LM algorithm to train the DNN network, but it often falls into a range of local minimum, resulting in the failure of the DNN model to converge or overfit during the training process. There are too many inconsistent factors for each UWA channel dataset label, which lead to this phenomenon, such as the uncertainty of the number of multipaths, the uncertainty of the multipath delay and the complex gain. These complex factors present challenges in the DNN model training. Table 3 lists the training parameters of the DNN network.

As shown in Fig. 6, the input data at the receiver end of the system is an unknown binary bit in the test stage that passes through the UWA channel. The received discrete probe signal $p_r(n)$ is extracted from the data frame by positioning the probe signal position, and then the DNN network completes the initial output value of the UWA channel impulse response, which can be expressed as:

$$p_r(n) = \sum_{l=0}^{L-1} p(n-l)h(l) + g(n), \quad n = 0, 1, \dots, N, \quad (25)$$

TABLE 2. The principle of the Rprop algorithm.

1: Initial parameter

The w_{ij} represents the weight between the i -th neuron of the l -th layer and the j -th neuron of the $l+1$ -th layer. The n_{ij} represents the learning rate corresponding to the weight w_{ij} . The $n_{\max}$ indicates the maximum learning rate. The $n_{\min}$ indicates the minimum learning rate.

2: Update rule

If the $\nabla J(w_{ij}^{(t)}) \times \nabla J(w_{ij}^{(t-1)}) > 0$, $n_{ij}^{(t)} = \min(n_{\max}, \delta^+ n_{ij}^{(t-1)})$;

If the $\nabla J(w_{ij}^{(t)}) \times \nabla J(w_{ij}^{(t-1)}) < 0$, $n_{ij}^{(t)} = \max(n_{\min}, \delta^- n_{ij}^{(t-1)})$;

If the $\nabla J(w_{ij}^{(t)}) \times \nabla J(w_{ij}^{(t-1)}) = 0$, $n_{ij}^{(t)} = n_{ij}^{(t-1)}$,

where, $0 < \delta^- < 1 < \delta^+$, The $\nabla J(w_{ij}^{(t)})$ and $n_{ij}^{(t)}$ respectively represent the gradient and learning rate at the time t . The $\nabla J(w_{ij}^{(t-1)})$ and $n_{ij}^{(t-1)}$ respectively represent the gradient and learning rate at the time $t-1$.

3: Weight update

$$w_{ij}^{(t+1)} = w_{ij}^{(t)} - n_{ij}^{(t)} \cdot \text{sign}(\nabla J(w_{ij}^{(t)})),$$

where, the $w_{ij}^{(t+1)}$ is the weight at the time $t+1$. The $w_{ij}^{(t)}$ indicates the weight at the time t . The $\text{sign}(\cdot)$ can judge the direction of the gradient function. If the $\nabla J(w_{ij}^{(t)}) > 0$, there will be the $\text{sign}(\nabla J(w_{ij}^{(t)})) = 1$. At the same time, if the $\nabla J(w_{ij}^{(t)}) < 0$, there will be the $\text{sign}(\nabla J(w_{ij}^{(t)})) = -1$.

TABLE 3. Training parameters for the DNN network.

Parameter	Value
Maximum number of epochs	8000
Performance goal of training	10^{-7}
Minimum performance gradient	10^{-8}
Minimum learning rate	10^{-6}
Training function	Rprop
Maximum validation failures	6

where $g(n)$ represents the additive white gaussian noise, $p(n)$ represents the OFDM modulated probe signal, N represents the length of the probe signal $p(n)$, and L represents the length of the UWA channel. The matrix form of equation (25) is:

$$\begin{pmatrix} p_r(0) \\ p_r(1) \\ \vdots \\ p_r(N-1) \end{pmatrix} = \begin{pmatrix} p(0) & 0 & \dots & 0 \\ p(1) & p(0) & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ p(N-1) & p(N-2) & \dots & 0 \end{pmatrix} \begin{pmatrix} h(0) \\ h(1) \\ \vdots \\ h(L) \end{pmatrix} + \begin{pmatrix} g(0) \\ g(1) \\ \vdots \\ g(N-1) \end{pmatrix}. \quad (26)$$

there is not the overfitting phenomenon in the testing and validation phases, and the generalization ability of the DNN model does not decrease for the shallow sea environment.

III. PARAMETER SETTINGS

A. UWA ENVIRONMENT PARAMETER SETTING

To improve the reliability and accuracy of the experiment, this study used the BELLHOP ray model to simulate a shallow sea environment [27], [28] to generate the UWA channel dataset and the original UWA test channel. The average depth of the water body is 20 m, and the water density is 1.0 g/cm^3 . In the sediment, the speed of sound is 2000 m/s, the attenuation coefficient is $0.5 \text{ dB}/\lambda$, and the density of the sediment layer is 1.8 g/cm^3 . The parameter settings of the transceiver in a shallow sea environment are listed in table 4.

TABLE 4. Simulation parameters of the transceiver.

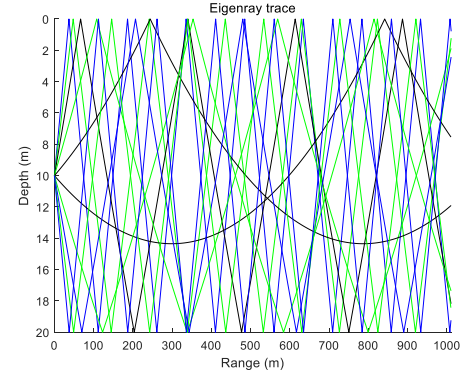
Parameter	Value
Sound frequency	1024 Hz
Horizontal distance	1000 m
Depth of transmitter	10 m
Depth of receiver	10 m
Angle of emergence	$-15^\circ \sim 15^\circ$

Based on the shallow sea environment and the parameter settings of the transceiver, the Bellhop model simulates the communication scene under the shallow water environment condition to generate the eigenray diagram and propagation loss diagram for the acoustic source depth of 10 m, as shown in Figs. 8-(a) and 8-(b), respectively. From the eigenray diagram shown in Fig. 8-(a), it can be seen that the propagation path of a sound wave is diverse, which will cause differences in the arrival delay of the signal. In Fig. 8-(b), the propagation loss shows that under different depth and range conditions, the signal strength will exhibit various degrees of attenuation during the propagation of the acoustic signal.

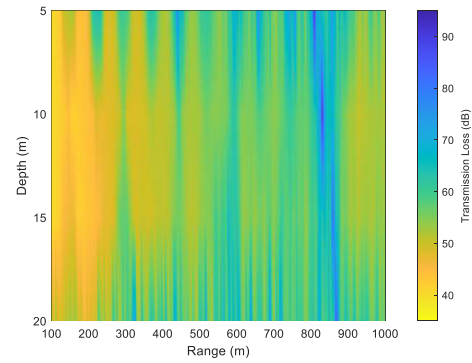
B. UWA TEST CHANNEL AND DNN-VTRM SYSTEM SETTING

The original UWA test channel is shown in Fig. 9-(a); however, it cannot be directly used in the DNN time domain channel estimator. The original UWA channel requires the high-resolution sampling, takes the geometric mean and the arithmetic mean of the phases for the same sampling delays, and normalizes the sampled results to obtain the normalized impulse response $h(n)$ as the UWA test channel, as shown in Fig. 9-(b). The number of multipath components of the $h(n)$ is five by the data processing.

To verify the influence of the proposed DNN-VTRM algorithm on the OFDM underwater acoustic communication system under the range of SNRs from 0 dB to 40 dB condition, the simulation processing does not consider the Doppler effect and Peak-to-Average Power Ratio (PAPR)



(a) Eigenray diagram



(b) Propagation loss

FIGURE 8. Eigenray diagram and propagation loss of the UWA channel.

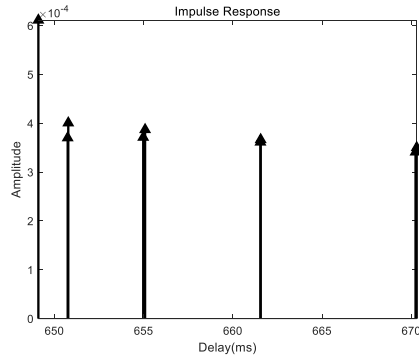
issues temporarily. The UWA test channel is regarded as a time-invariant UWA channel within an OFDM symbol period. The parameters of the OFDM underwater acoustic communication system based on DNN-VTRM are listed in table 5.

IV. SIMULATION EXPERIMENTS AND ANALYSIS

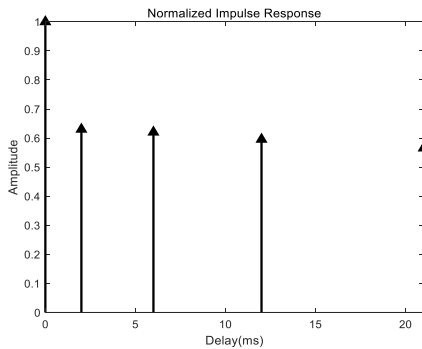
A. UWA VTRM CHANNEL ANALYSIS

By applying the VTRM process to the actual channel $h(n)$, it can obtain the actual virtual time reverse channel $h'(n)$, as shown in Fig. 10. And Fig. 10-(a) shows the actual channel $h(n)$, and Fig. 10-(b) shows the actual virtual time reversal channel $h'(n)$ ($h'(n)$ is the convolution of $h(n)$ with $h(-n)$).

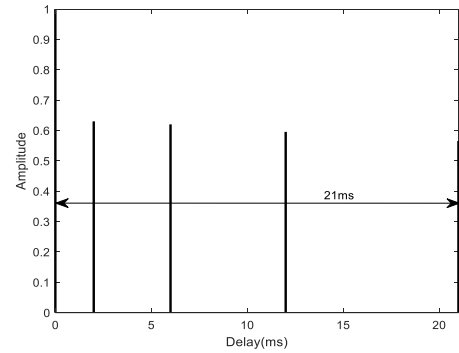
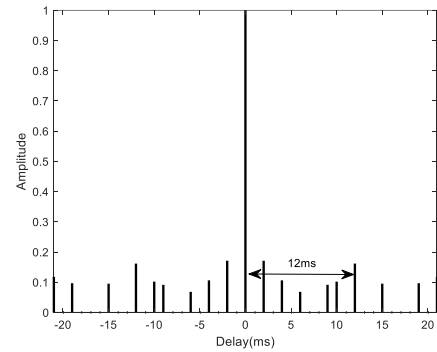
In Fig. 10-(b), the virtual time reversal channel $h'(n)$ has symmetry and the main peak $h'(0)$ concentrates all multipath components, which can significantly suppress the ISI caused by the multipath expansion in underwater acoustic communication system. Meanwhile, the component of other multipath is significantly less than the main peak, and some multipath component of the $h'(n)$ is less than a certain threshold that can be regarded as noise. The main multipath component of the $h(n)$ is concentrated in the range from 0 ms to 21 ms in Fig. 10-(a). However, the main multipath component of the $h'(n)$ is concentrated in the range from 0 ms to 12 ms, in Fig. 10-(b). In addition, the multipath component of the



(a) Channel impulse response



(b) Normalized impulse response

FIGURE 9. Channel impulse response and normalized impulse response.(a) Original UWA channel $h(n)$ (b) Original UWA VTRM channel $h'(n)$ **FIGURE 10.** Original UWA channel and VTRM channel.**TABLE 5.** Simulation parameters of the DNN-VTRM system.

Parameter	Value
Sampling rate	48 KHz
Bandwidth	4 KHz
Number of subcarriers	512
Type of modulation	16/64QAM
Probe signal length	64 ms
Cyclic prefix length	24 ms
Cyclic suffix length	24 ms
Guard interval	24 ms
Channel model	UWA sparse multipath channel
Noise model	Additive White Gaussian Noise

$h'(n)$ in the range from 12 ms to 21 ms is significantly smaller than the multipath component in the range from 0 ms to 12 ms, which can be regarded as noise. The amplitude ratio of the side lobe to the main lobe of the $h'(n)$ is much lower than that of the original UWA channel $h(n)$, which compresses the time length of the UWA channel.

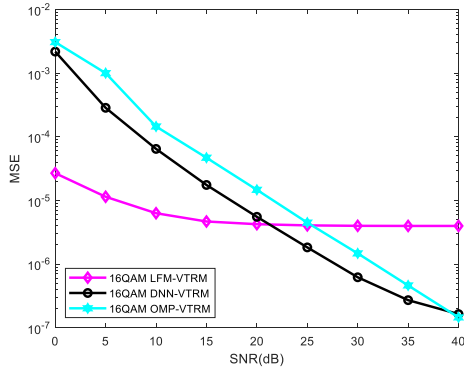
The side lobe of the $h'(n)$ still has some multipath components that will cause the ISI to the received symbols, as shown in equation (10). By adding a cyclic suffix of the same length

as the cyclic prefix that is larger than the maximum delay length of the UWA test channel, the multipath extension caused by these multipath components is overcome, and the received symbol shown in equation (14) is finally obtained.

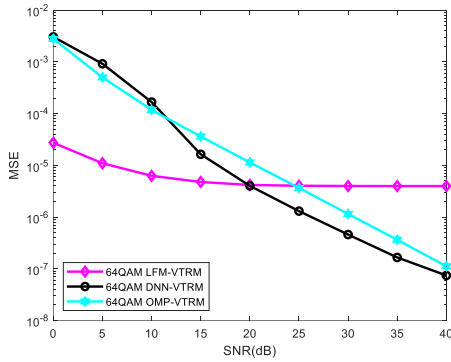
B. DNN-VTRM SYSTEM MSE ANALYSIS

The simulation experiment uses the OFDM system based on the DNN-VTRM, as shown in Fig. 1, to verify the MSE performance with different time domain estimation methods and the BER performance for different VTRM systems. In the [29], it is proposed that the LFM signal, which exhibits sharp autocorrelation peaks, be used to complete the channel estimation step of the VTRM. The channel estimation principle of the LFM signal is based on matched filtering. In the [30], the OMP algorithm is used to complete time domain channel estimation. In this section, the LFM matched filtering algorithm and the OMP algorithm are used to complete the time domain channel estimation of the UWA channel $h(n)$ as a contrast. The OMP algorithm needs to know the sparsity of the $h(n)$ in advance, but LFM matching filter algorithm and DNN time domain estimator do not need. Therefore, these two methods are more practical.

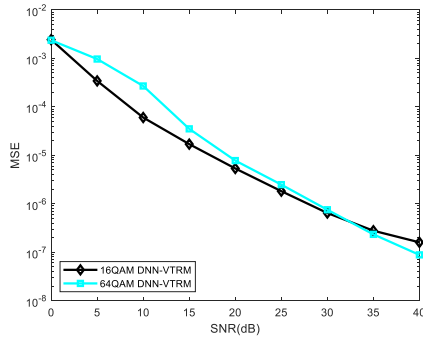
Fig. 11-(a) shows the MSE curves estimated by the LFM matched filtering algorithm and the DNN time domain estimator for the UWA test channel $h(n)$ under the 16QAM



(a) MSE for LFM/OMP/DNN under 16QAM modulation



(b) MSE for LFM/OMP/DNN under 64QAM modulation



(c) MSE for DNN under 16QAM and 64QAM modulation

FIGURE 11. MSE performance of LFM/OMP/DNN channel estimation.

modulation condition. Fig. 11-(b) illustrates the MSE curves estimated by the LFM matched filtering algorithm and the DNN time domain estimator for the UWA test channel $h(n)$ under the 64QAM modulation condition. As can be seen in Fig. 11-(c), the MSE curve is estimated by the DNN time domain estimator for the UWA test channel $h(n)$ with 16QAM and 64QAM modulation.

From Figs. 11-(a) and 11-(b), as the signal-to-noise ratio (SNR) increasing, there will be an obvious floor effect that the MSE accuracy of the LFM matched filtering algorithm is not improved. Under the 16QAM and 64QAM modulation condition, the DNN time domain estimator and the OMP algorithm can solve the above problem, but the DNN time domain

estimator is more accurate than the OMP algorithm. Since the probe signal $p(n)$ adopts different modulation modes to perform the time domain estimation task for the DNN, it can be seen from Fig. 11-(c) that the MSE performance under the 16QAM modulation condition is slightly better than that under the 64QAM modulation condition, but there is no obvious gap among different modulation methods for the DNN time domain estimator. The Fig. 11-(c) demonstrates that the probe signal $p(n)$ does not have a significant impact on the accuracy of the DNN time domain estimator under different modulation schemes.

C. DNN-VTRM SYSTEM BER ANALYSIS

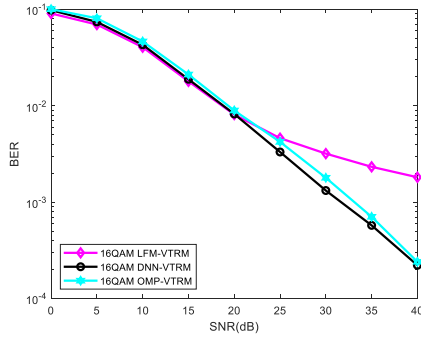
When the time domain channel estimation is completed by the LFM matched filtering algorithm, the OMP algorithm, and the DNN time domain estimator, the BER performance of the traditional VTRM system will be compared and analyzed for the three UWA channel estimation methods. Figs. 12-(a) and 12-(b) show the BER of the LFM-VTRM, the OMP-VTRM, and the DNN-VTRM system under the 16QAM and 64QAM modulation condition, respectively. Fig. 12-(c) shows the BER performance of the DNN-VTRM system under the 16QAM and 64QAM modulation.

From the analysis of the MSE performance in the Fig. 11-(a) and 11-(b), when SNR is less than 20 dB, it can be seen that the estimation accuracy of the LFM matched filtering algorithm is higher than that of the OMP algorithm and the DNN time domain estimator. When the SNR is greater than 20 dB, the channel estimation accuracy of the DNN time domain estimator and the OMP algorithm is higher than that of the LFM matched filtering algorithm.

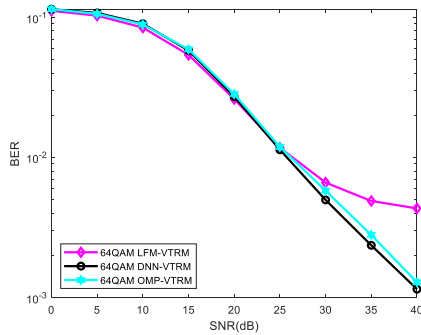
However, Figs. 12-(a) and 12-(b) show that when the SNR is less than 20 dB, the BER performance of the LFM-VTRM, OMP-VTRM, and the DNN-VTRM system are similar, because the DNN time domain estimator and the OMP algorithm have insufficient estimation accuracy, but the VTRM can improve system gain under the low SNR condition. Therefore, the performance of the DNN-VTRM and OMP-VTRM system are not significantly different from the LFM-VTRM system.

In addition, the performance of the VTRM also depends on the accuracy of the channel estimation, and the floor effect of the LFM matched filtering algorithm will eventually do not make the BER performance enhancement of the LFM-VTRM system than before. Figs. 12-(a) and 12-(b) show that the BER curve of the LFM-VTRM system will appear obvious floor effect, but the DNN-VTRM and OMP-VTRM systems perform better than the LFM-VTRM system under the high SNR condition. The accuracy of the DNN time domain estimator is higher than that of OMP algorithm, as shown in Figs. 11-(a) and 11-(b). Therefore, the BER performance of the DNN-VTRM system is better than that of the OMP-VTRM system, as shown in Figs. 12-(a) and 12-(b).

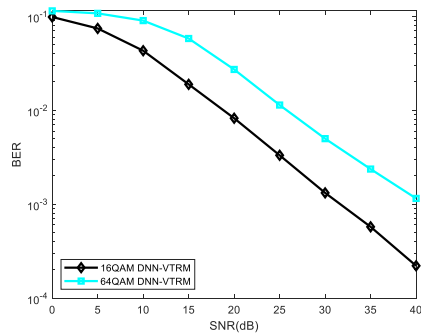
As shown in Fig. 12-(c), the DNN-VTRM system always satisfies that the BER performance of the 16QAM modulation is better than that of the 64QAM modulation. In addition,



(a) BER for LFM-VTRM/OMP-VTRM/DNN-VTRM under 16QAM modulation



(b) BER for LFM-VTRM/OMP-VTRM/DNN-VTRM under 64QAM modulation



(c) BER for DNN-VTRM under 16QAM and 64QAM modulation

FIGURE 12. BER performance of LFM-VTRM/OMP_VTRM/DNN-VTRM system.

combined with Fig. 11-(c), it can be seen that the time domain estimation results of the probe signal $p(n)$ under the different modulation schemes are slightly different, but this difference does not affect the BER performance of the system under the different modulation schemes.

D. COMPARATIVE ANALYSIS OF CHANNEL ESTIMATION

From the above conclusion, the accuracy of the DNN time domain estimator is the higher and the equalization effect of the DNN-VTRM system is the better. This study uses the BER as a criterion to measure different communication systems. In order to compare and analyze the performance of the proposed DNN-VTRM system, which is compared

TABLE 6. Comparison of the running complexity and the running time performance for different channel estimation methods.

Method	Complexity	Runtime (ms)
LS	N	0.2
LMMSE	N^2	2
DNN	$5N$	8.2
LFM-VTRM	—	0.7
OMP-VTRM	KPM	1.6
DNN-VTRM	$5N$	9.3

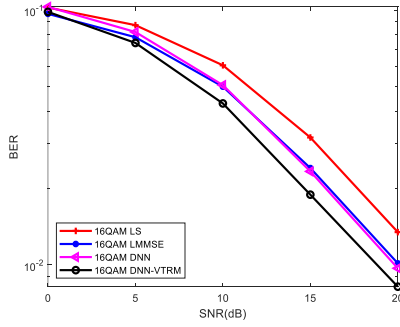
with the LS algorithm, the LMMSE algorithm, and the DNN frequency domain estimator without the VTRM processing under the 16QAM and 64QAM modulation condition, respectively. Figs. 13-(a) and 13-(b) show the BER simulation result under the low SNR range (from 0 dB to 20 dB), and Figs. 13-(c) and 13-(d) show the BER simulation results over the entire SNR range (from 0 dB to 40 dB). Figs. 13-(e) and 13-(f) show the MSE performance results of the LS algorithm, the LMMSE algorithm, and the DNN frequency domain estimator without the VTRM processing, respectively.

As can be seen from Figs. 13-(e) and 13-(f), the accuracy of the DNN frequency domain estimator is close to the LMMSE algorithm and better than the LS algorithm in the range of the low SNR (from 0 dB to 20 dB). However, the DNN frequency domain estimator is better than the LS and LMMSE algorithms in the high SNR range (from 20 dB to 40 dB). Either the 16QAM or 64QAM modulation, the BER performance of the DNN frequency domain estimator without the VTRM is better than the LS algorithm, close to or better than the LMMSE algorithm, but worse than the DNN-VTRM system.

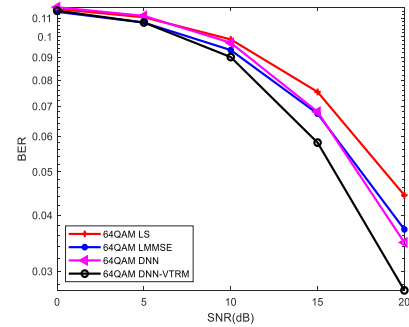
Under the premise of the UWA channel accurate estimations, the characteristic of the VTRM focusing channel energy can improve the system gain under the low SNR condition. The BER performance of the DNN-VTRM system is significantly better than the LS algorithm, the LMMSE algorithm, and the DNN frequency domain estimator without the VTRM processing with a low SNR range (from 0 dB to 20 dB), as shown in Figs. 13-(a) and 13-(b). The DNN-VTRM can improve the BER performance of the OFDM system at the low SNR condition.

In Figs. 13-(c) and 13-(d), the BER performance of the DNN-VTRM system is significantly better than that of the LS, the LMMSE, and the DNN frequency domain estimator at a high SNR range (from 20 dB to 40 dB), regardless of the 16QAM or 64QAM modulation.

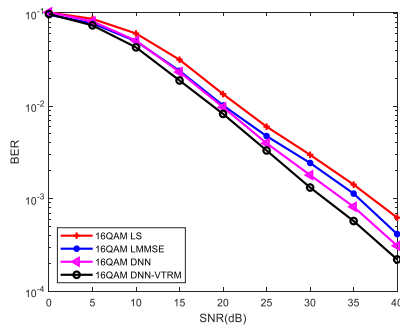
In summary, due to the high channel estimation accuracy of the DNN channel estimator and the focused channel energy characteristics of the VTRM, the DNN-VTRM system is better than traditional estimation algorithms under the low SNR condition.



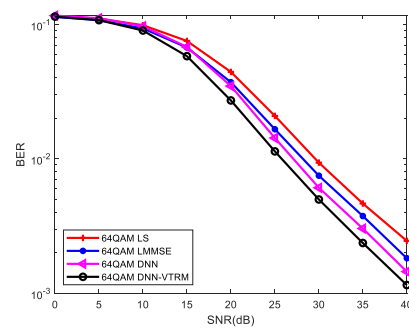
(a) BER for LS/LMMSE/DNN/DNN-VTRM under 16QAM modulation at low SNR



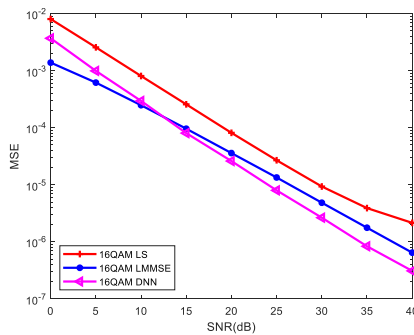
(b) BER for LS/LMMSE/DNN/DNN-VTRM under 64QAM modulation at low SNR



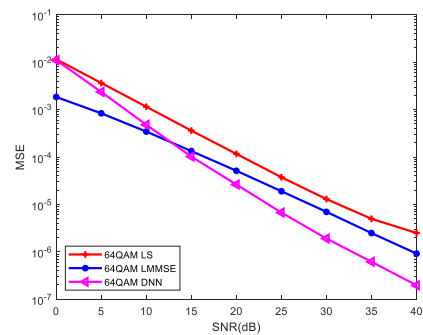
(c) BER for LS/LMMSE/DNN/DNN-VTRM under 16QAM modulation



(d) BER for LS/LMMSE/DNN/DNN-VTRM under 64QAM modulation



(e) MSE for LS/LMMSE/DNN under 16QAM modulation



(f) MSE for LS/LMMSE/DNN under 64QAM modulation

FIGURE 13. Comparison of BER and MSE performance with various methods, each of which is modulated with 16QAM and 64QAM, respectively.

E. COMPARISON OF THE COMPLEXITY AND REAL-TIME PERFORMANCE

Table 6 lists the running complexity and the running time of the proposed DNN-VTRM algorithm with other methods in an OFDM symbol. The running complexity is the number of times the calculation algorithm is multiplied.

The DNN network has 5 layers structure with three hidden layers. Each layer of the DNN does the addition and multiplication, so the running complexity of each layer is the same as the LS algorithm. The OMP-VTRM algorithm requires PM product times in each iteration and requires through K iterations. The total product time required by the

entire algorithm is KPM , where K is related to the channel sparsity, P represents the number of atoms in the dictionary, and M represents the dimension of each atom.

The experiments are performed in MATLAB R2023a using an Intel Core i7-10700 CPU and 16 GB of memory storage for calculation. table 6 shows that the LS and the LFM-VTRM algorithm take the least time, followed by the LMMSE and the OMP-VTRM algorithm. The DNN-VTRM and the DNN frequency domain estimator require the most running time. The DNN frequency domain estimator and DNN-VTRM use the same neural network structure, but the DNN-VTRM must complete an additional virtual time reversal processing step

that is a complex process. So, the calculation time of the DNN-VTRM is longer than that required by DNN frequency domain estimator.

V. CONCLUSION

Aiming at the OFDM underwater acoustic communication system, we propose a new channel estimation and equalization algorithm based on the DNN and VTRM technologies to solve the residual ISI problem caused by the UWA channel and nonlinear interference in the channel estimation. The DNN time domain estimator learns the nonlinear characteristics of the UWA channel by a large number of UWA datasets, so that the DNN time domain estimator can provide accurate delay, complex gain and phase information for the channel estimation. It can solve the nonlinear interference problem of the UWA channel and improve the equalization performance of the VTRM. Simulation experiments and research results show that the proposed algorithm can compress the channel time length and focus the channel energy through the virtual time inverse technology, which can better solve the residual ISI in OFDM underwater acoustic communication system.

At the same time, the deep neural network can further solve the nonlinear effect of complex factors on the UWA channel estimation. Moreover, the DNN-VTRM system does not need the pilot data for the channel estimation, nor does it need an LFM signal to complete the matched filtering in the traditional VTRM algorithm. It can not only solve the floor effect problem of the LFM matching filtering algorithm, but also greatly improve the effectiveness of the communication system. In the system reliability, compared with the OMP-VTRM algorithm based on compressed sensing, the DNN-VTRM has a higher time domain estimation accuracy and outstanding BER performance. The combination of the DNN time domain estimator and the VTRM is better than the DNN frequency domain estimator without the VTRM processing and has an excellent BER performance than the traditional LS and LMMSEs frequency domain estimation algorithms. In conclusion, the UWA channel estimation and equalization algorithms based on the DNN and VTRM have important theoretical and application value for improving the OFDM underwater acoustic communication system.

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